

Automated Sugarcane Disease Detection Using Faster RCNN with an Android Application

Dr.R.Murugeswari

Department of Computer Science and EngineeringKalasalingam
Academy Of Research and EducationAnand Nagar,
Krishnankovil, Tamilnadu, India
r.murugeswari@klu.ac.in

V.Raja Dhananjeyan

Department of Computer Science and EngineeringKalasalingam
Academy Of Research and EducationAnand Nagar,
Krishnankovil, Tamilnadu, India
itzurdhananj@gmail.com

Z.Sharik Anwar

Department of Computer Science and EngineeringKalasalingam
Academy Of Research and EducationAnand Nagar,
Krishnankovil, Tamilnadu, India
rawnakirahs327@gmail.com

C.Naveen Karthik

Department of Computer Science and EngineeringKalasalingam
Academy Of Research and EducationAnand Nagar,
Krishnankovil, Tamilnadu, India
naveenkarthik1207@gmail.com

Abstract—Sugarcane is a crop that is grown vastly in our country. Diseases are common in plants and so in sugarcane as well. Visual identification of sugarcane diseases is the method commonly used by a farmer to detect and identify them. This technique takes more time and becomes difficult in this process in huge farms. If not treated earlier, it causes a threat both to the farmers and the sugarcane industry. Detecting them in early stages using Deep Convolutional Neural Network can prevent losses. This work attempts to combine Convolutional Neural Network architectures of Faster Region-based Convolutional Neural Network to improve detection and recognition of sugarcane illnesses.1500 pictures of healthy sugarcane leaves and afflicted sugarcane illnesses were used to train the models. An android application also serves as the end user interface for capturing photographs and quickly determining the disease's and it's accuracy.

Index Terms—sugarcane diseases, Convolutional Neural Network, Faster Region-based Convolutional Neural Network, Residual Neural Network.

I. INTRODUCTION

Sugarcane and sugar utilization have a long and storied history in the country's socioeconomic landscape.. The present-day rural economy set up sugarcane cultivation and the sugar industry has been a focal point for economic development in the rural areas. This will be affected if the crop's cultivation and harvest is reduced because of the diseases that attack them. Traditionally, diseases in plants were identified by experienced farmers or agricultural scientists in laboratories. The latter can be an highly-priced technique of understanding the disease inflamed at the plant while, the previous can be faulty and vulnerable to human errors. Hence to help the farmers detect the diseases easily and to increase the development to make agriculture less complicated is the most important intention of this paper.

A. Types Of Sugarcane Diseases

- **Rust Disease** In rust disease, elongated yellowish patches seen on both surfaces of the leaves are the first signs of common rust. These spots grow in size, mostly long, and change color from red-brown to brown. Around the lesions, a narrow pale yellow-green halo appears. Rust causes various lesions, the most of which are serious. Individual leaves have a brown or rust color in general.

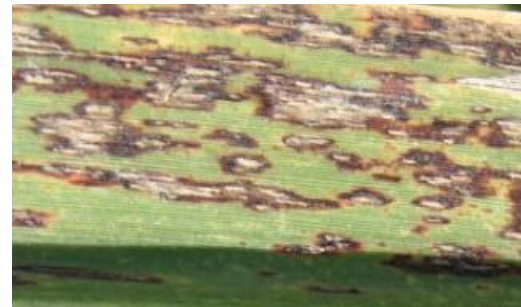


Fig. 1. Rust Disease.

- **Yellow Leaf Disease** A yellowing of the leaf midrib on the underside of the leaf is a symptom of SCYLD. Starting with the tallest growing spindle leaf, the yellowing begins on leaves 3 to 6. Yellowing of the lower surface of the leaf midrib on leaves 3 to 6, starting from the highest extending spindle leaf, are the first signs of yellow leaf.



Fig. 2. Yellow Leaf Disease.

- **Red Rot Disease** Drying may be seen on the spindle leaves (3rd 14th). The stalks get discoloured and hollow as time goes on. Acervuli is a black fruitingbody that

develops on nodes and/or rind. When the disease stalk is split, a sour odour is released. Inside the tissues, there are interwoven transverse white dots that are reddish.



Fig. 3. Red Rot Disease.

- **Eye Spot Disease** Sugarcane eyespot and seedling blight are caused by a fungal disease. Spots on the leaves and stalks run parallel to the veins and are elliptical in shape, measuring 12 mm in diameter and reddish with golden haloes. The above-mentioned dots range in colour from reddish brown to golden brown. They reach towards the tips of the leaves and stretch upward.



Fig. 4. Eye Spot Disease.

II. LITERATURE REVIEW

K. Renugambal et al. [1] has proposed an artificial intelligence fashion auto-discovery and sugarcane bracket splint conditions based on image processing have been proposed. Images of diseased leaf sugarcane are taken/captured by the camera. To reduce neighbour infection of leaves, pre-process the image using histogram image equalisation, filter, color-changing, and/or segment methods. Using the Support Vector Machine classifier, the infected splint was also employed for the bracket. This might be accomplished by using the K means clustering approach to join the splint area with the impacted area in the diseased neighbourhood of a splint. Finally, the impacted area's GLCM texture point is calculated to classify the situations. Both point analysis and segmentation fashions were used on various infected sugarcane shops during the development of this technique.

Evy Kamilah Ratnasari et al. [2] has proposed that this disease detection approach is based on a work on a digital image to determine the type of spot illness supported by a classified model. Rice disease and infection identification is achieved with the help of texture analysis (TA) utilising descriptor fractal and "S" 4 color of space of HSV as a component with a minimum of 83 percent and infection leaf picture categorization. Technique categorize is an add-on that will not prevent fungal disease on sugarcane plants. With a precision of 95.25 percent, the determined segment location of a disease image.

K. Kavitha et al. [3] has proposed some image processing techniques to detect sugar cane disease, the system collects leaf images, processes them using adaptive histogram equalization (AHE), and uses the k-means clustering algorithm for segmentation. It will be replaced, according to the document. Gray Level Co-occurrence Matrix (GLCM) and Principal Component Analysis are used to extract statistical variables such as variance, skewness, standard deviation, mean, and covariance (PCA). Ultimately, the creator of the project decided on (SVM) for detection and classification with 95% average accuracy.

Shima Ramesh et al. [4] has proposed using Random Forest in relation between diseased and healthy splint from the information created the sets during the article. That study includes a colourful depiction of the perpetration phase videlicet generate data set, point birth, bracket, and or classifier. The illness and health plant part leaf information is combined with Random-Forest training to provide a more accurate image depiction of health and disease. Histogram-Acquainted-Grade was used for the image's features router. The use of an overall machine to educate a large set of data that was proven to be intimately transparent resulted in a complaint owing to a decrease in massive scales of a major shop.

Arpan Kumar et al. [5] The colour degradation analysis of diseased leaves or plants, as shown in their study utilising Processing Image, is an important step in manual identification of the disease and thus the type of disease. Soft computing approaches are frequently used to build expert systems and are crucial in the development of knowledge-based systems. Farmers will be ready to use this technology to find answers to their farming problems. Visual tools for disease detection are created by analysing the leaves of sugarcane plants.

Sammy Militante et al. [6] has proposed in their research that one of the issues in the sugar assiduity is sugarcane conditions that produce increased growth plants overran with complaint performance within the fiscal small scale farmers are conditions that aren't treated detecting earlier. The employment of depth literacy methods by a machine to study vision provides a solution to this challenge. This result was graded and tested in a shallow literate sugarcane sample predicting the data sets of complaint in damaged leaves and undamaged (health) leaves, yielding a 95 percent accuracy. His working model was trained to detect and categorise infected and un diseased sugarcane leaves in images. As a result, this work proposes a paradigm for assisting growers in detecting and classifying pests using deep literacy methods.

Melike Sardogan et al. [7] Plant Leaf Disease Detection and Classification Using Convolutional Neural Network and the LVQ Algorithm has been proposed. The importance of early detection of problems cannot be overstated. After detecting the symptoms of splint problems, automatic styles assist in taking action. For the detection and bracketing of tomato splint complaints, this research provides a Convolutional Neural (CNN) model and Learning Vector Quantization (LVQ) algorithm grounded system. With four symptoms, 500 images of leaves is present in the dataset. For automatic point birth and bracketing, we've created a Convolutional Neural Network model. For factory splint complaint research, colour information is painstakingly used. The contaminants are applied to three channels based on RGB factors in our model. For training the network, the LVQ was fed the affair point vector of the complication section. The results of the experiments show that the suggested system can distinguish four different types of tomato splints.

Snehal Pawar et al. [8] has presented a model to find out kinds of conditions, but the three most common are rust spots, unheroic spots, and ring spots. Although the complaint can be identified with the naked eye, the identification method will provide the best-estimated outcome. The current technique employs a digital image of a splint to describe the discomfort, but not everyone will be able to obtain a high-detail digital image. Using a retired Markov model and an an isotropic prolixity method, we proposed a new approach to describe and classify the complaint. The stoner will be able to use a mobile camera to capture an image, and the image will be able to be named from the gallery.

III. METHODOLOGY

DATA SET ACQUISITION: Images in the data set for our project is captured manually handpicked directly from the fields using different modes of camera and natural lighting and it is saved in a separate respective folder distinguishing the diseased images. It is beneficial to obtain a more precise categorization. A new class was introduced to the data set to identify healthy leaves from unhealthy ones. There are only images of healthy leaves in there. The data set of our project consists of 1500 images including 4 important types of sugarcane leaf diseases Eye spot, red rot, White rust and Yellow leaf. Each image in the data set is stored in uncompressed JPG format or PNG format.



Fig 5. Dataset Sample images

IMAGE AUGMENTATION AND PRE-PROCESSING:

The primary objective of using augmentation is to expand the data set size and provide minor deformation to the images. It aids in the prevention of overfitting during the training procedure. The image augmentation includes one or more transformation procedures, such as flipping, rotating, and so on. pre-processing are keras functions that are used to produce and export augmented multiple images to a folder in order to create a large data set of changed photos. Additionally, the image pre-processing method comprises manually cropping all of the images and squaring around leaves to highlight the area of interest.

TRAINING THE MODEL: We develop an image classification model using the data set and a deep convolutional neural network. Faster Region-based Convolutional Neural Network is one of many well-known state-of-the-art deep learning-based pre trained models that can be used in both research and industry. Deep learning models with thousands of variables (weights) frequently require a vast quantity of data and computational resources to train from beginning. Transfer learning is a method for speeding up this process by reusing a chunk of a model which has already been trained on a similar task in a new model.

- Conv2D is a convolutional layer that extracts features from an image or sections of an image. Pooling layers minimize the complexity of every characteristic so that the most significant parts are highlighted.
- Finally, there's a completely linked Dense layer that employs a flattened representation of the characteristics detected in the preceding layers to generate an image prediction.
- Finally, with the training of the classifier layer, train the weights of the upper layers of the previously trained model for better performance. The weights will be forced to be modified from general feature maps to features unique to the Sugarcane dataset throughout the training phase. The stored model is then created from the trained model.

DEPLOYMENT OF TRAINED MODEL: The trained model obtained in the form of a stored model is incompatible with Android application integration. The stored model is transformed into. tflite format using the TensorflowLite framework. It is converted so that the. tflite format can only be deployed into android application. TensorFlowLite is composed of two components. TensorflowLite interpreter and a TensorflowLite converter. The TensorFlowLiteconverter transforms TensorFlow models into an efficient version (.tflite) for use by the interpreter, using optimizations to reduce binary size and enhance speed. Based on the previously trained model analysis, the TensorFlowLite interpreter then runs specifically optimized models on mobile phones to detect and categorize the photographs into relevant illness categories.

DETECTION OF DISEASED SUGARCANE PLANTS:

The trained model has been linked with the android application at this point. The Apk is extracted and the app is installed in the mobile phone. When we point the diseased plant in front of the camera of the mobile phone, the input image is taken. The input photos are then analyzed using the trained model(which are bounded to the app as .tflite model) and categorized into a specific disease category, which is then provided to the end user via the app user interface.

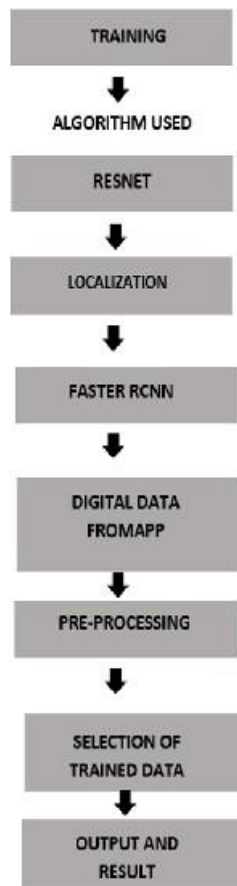


Fig. 6. Workflow Diagram.

In the advent of ai and automation processes, deep learning is critical. Deep Learning is made up of a large number of neural networks that are managed by the computer's processor or integrated video processor. In a neural network, each neuron is known as a single node. Deep learning has been used in a range of applications, including crop range classification and detection, plant identification and classification, and picture fruit grading. Photographs taken with a smartphone camera or another camera device attached to a robot have become increasingly popular. CNN, or Convolutional Neural Network, is gaining popularity among scientists.

There are multiple ways in today's world to combat the

problem of plant diseases, one of which is a Convolutional Neural Network. With the advent of deep learning in the identification of agriculture and plant diseases, it has recently become popular due to its performance and accuracy. Convolutional Neural Network 's are a deep learning technique that can handle difficult processes and perform image-based pattern recognition. Convolutional Neural Network is the most often used way for presenting sophisticated approaches and performing pattern recognition applications using a huge amount of data. We have created an android application to process the image through various popular architectures. We will be using three popular architectures. A machine learning algorithm that resides as a back end on our application and does automated inference for absence or presence of a disease. It and gives the output as the sugarcane leaf is diseased or healthy. Our android application captures the real-time image of infected sugarcane crop which is going to be our frontend. We trained our model using MakeSense and TensorFlow. Then we convert our trained model as tf lite file. This file can be used in our android application. The trained model can be imported into the android application and can be used for further classification and detection purposes. To accurately map the infected regions, faster Region-based Convolutional Neural Network will be used. The next step is localizing the infected regions. The models had been trained on sugarcane dataset. The trained models can be used to detect the disease successfully.

A. Algorithm

Faster-RCNN: A state-of-the-art detections network, Faster R-CNN, is employed for assessment to accurately recognize contaminated regions inside the images. To generate region of interest or infected region, it uses selective method. Faster-Region-based Convolutional Neural Network work with RPN which is Region Proposal Network. RPN may be a trained end-to-end model, that simultaneously predicts object boundaries and object scores at each detection.



Fig. 7. Annotated Leaf Image showing redrot

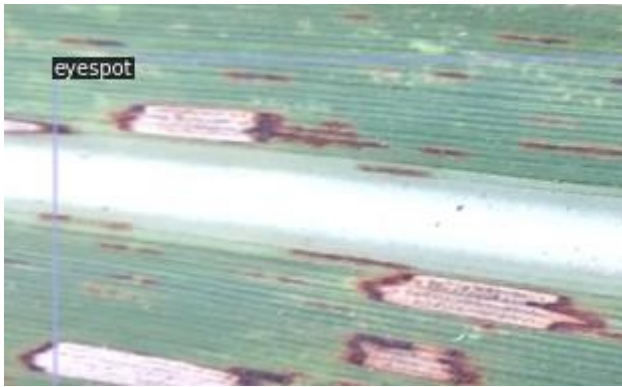


Fig. 8. Annotated Leaf Image showing eyespot



Fig. 9. Annotated Leaf Image showing healthy



Fig. 10. Android App Interface

A. Android Application

The android application that we have designed serves as the user-end interface for the farmers to take real time images of diseased leaves. The app has been designed using java in Android Studio, in such a way that any person can use the app without any hassle. The user needs to open his mobile camera and point it towards the leaf. After which the app displays the name(s) of the disease and the accuracy of the disease detected.

IV. RESULT AND CONCLUSION

Average Precision Mean As evident from the graphs given below, fig[6] shows higher mean average precision of the proposed algorithm(Faster RCNN) compared to the other model YOLOV5.

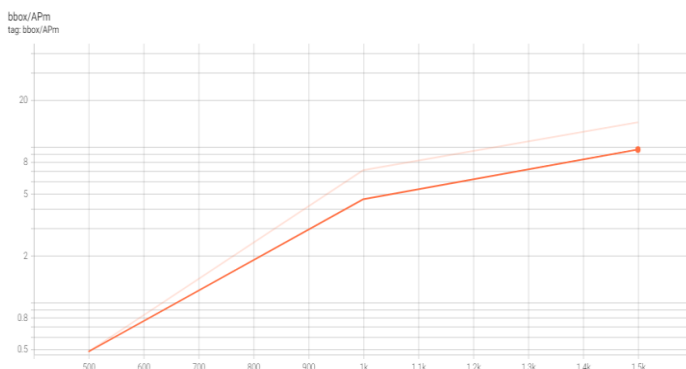


Fig 11. Average Precision Mean for Faster RCNN and YOLOV5

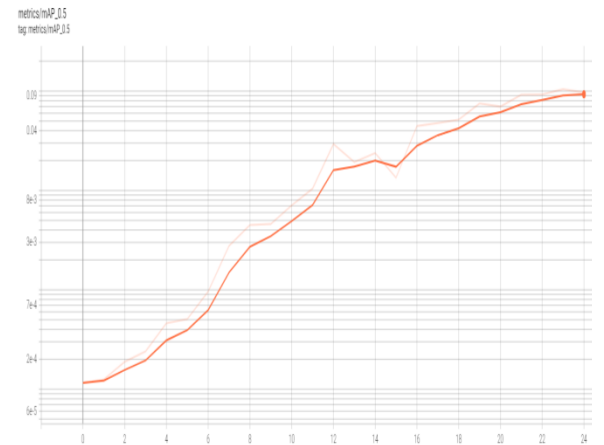


Fig 12. Average Mean Precision of YoloV5 Algorithm

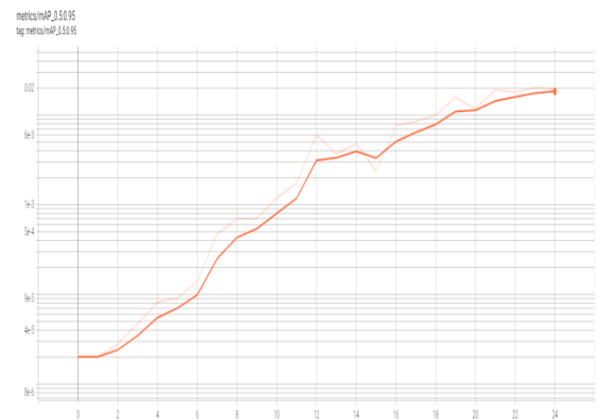


Fig 13. Average Mean Precision of YoloV5 Algorithm

Learning Rate

A comparison is made with the below learning rate graphs from the two models of Faster RCNN and YoloV5. The learning rate graphs of the YoloV5 models show a decline after a certain period whereas the Faster RCNN model is high and does not decline.

With all the factors taken into consideration, Faster RCNN proves to be a better and efficient algorithm in detecting the diseases compared to the YOLO V5 algorithm and this will help the farmers in predicting the diseases more accurately.

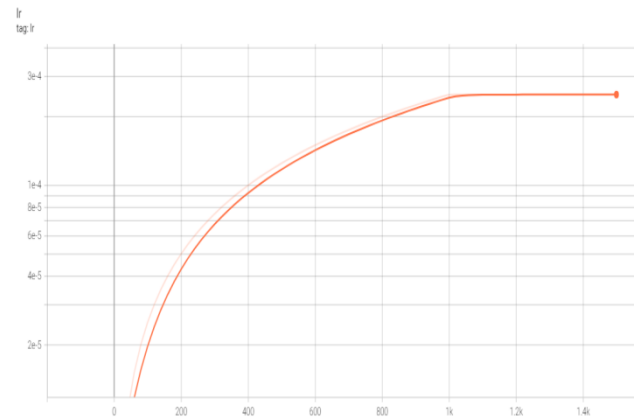
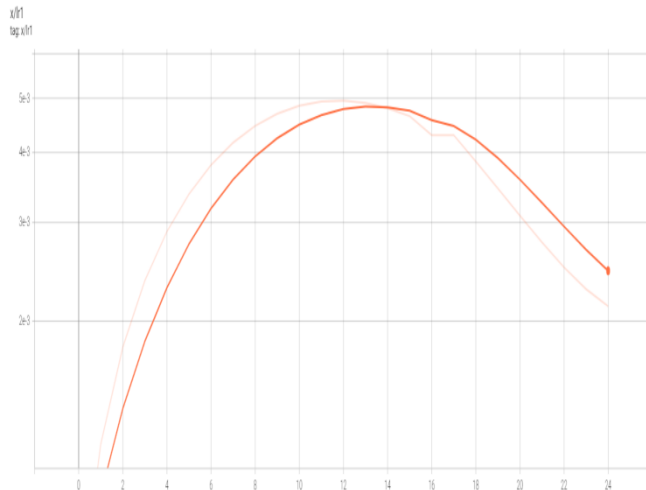


Fig. 14. Learning Rate of Faster RCNN Algorithm

Fig. 15. Learning Rate of YoloV5 Algorithm

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